

Advanced Financial Management (AFM) December 2022 Examiner's report

The examining team share their observations from the marking process to highlight strengths and weaknesses in candidates' performance, and to offer constructive advice for those sitting the exam in the future.

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General comments

This examiner's report should be used in conjunction with the published December 2022 sample exam which can be found on the [ACCA Practice Platform](#).

In this report, the examining team provide constructive guidance on how to answer the questions whilst sharing their observations from the marking process, highlighting the strengths and weaknesses of candidates who attempted these questions. Future candidates can use this examiner's report as part of their exam preparation, attempting question practice on the [ACCA Practice Platform](#) and reviewing the published answers alongside this report.

Format of exam

The examination comprised two sections, A and B. Section A consisted of one compulsory question for 50 marks in total. Section B consisted of two compulsory questions for 25 marks each. Out of this total of 100 marks across sections A and B, 20 marks were available for professional skills related to communication, commercial acumen, analysis and evaluation. 80 technical marks were available for applying appropriate technical knowledge in response to the requirements.

Section A

Question 1 – Fondir Co

The screenshot shows the ACCA exam interface for Question 1 – Fondir Co. The interface is divided into a left-hand sidebar and a main content area. The sidebar contains sections for Exhibits, Requirements, and Response Options. The main content area displays the question text and the list of exhibits.

Exhibits

- 1. Fondir Co, risk management
- 2. Receipt from Lothil
- 3. Short-term investment

Requirements

- Requirements (50 marks)

Response Options

- Word Processor
- Spreadsheet

The following exhibits, available on the left-hand side of the screen, provide information relevant to the question:

1. Fondir Co, risk management - includes details on communication
2. Receipt from Lothil
3. Short-term investment

This information should be used to answer the question requirements within your chosen response option(s).

Format of the question

Fondir Co is mainly drawn from section E of the syllabus, focusing on advanced risk management techniques. The individual question parts are broadly similar in scope and complexity to requirements that have appeared in previous risk management questions and should not have been unexpected. The requirements tested the ability of candidates to undertake a number of detailed calculations and to discuss issues related to those calculations and other associated points.

Comment on time management

Fondir Co is worth 50 marks which translates into 90 minutes for candidates to spend on this question (based on 1.8 minutes per mark). It is important that candidates thoroughly read the question to become familiar with the scenario. Candidates must then plan their time to have the opportunity to gain the maximum marks for each requirement.

Candidates should avoid the temptation to expand their answer to one requirement beyond the time allotted, using time that should have been spent answering the remaining requirements and the other questions in the exam. Markers cannot allocate more marks to a requirement than is available, even if the comments may otherwise be worthy of marks.

Scenario background

Exhibit one presented information about a food production company, Fondir Co, which is based in the USA. Due to rapid growth, the company is expanding its production facilities and earning revenue in a variety of currencies. The company's board is considering ways in which to manage its financial risk from exchange rate and interest rate fluctuations although the finance director argued that some risks were more difficult to manage than others. The exhibit provided an example of such a scenario based on Fondir Co's experience in the Italian market, where it has experienced a decline in revenue due to the weakening of the euro against the dollar. Exhibit two presented information about how foreign exchange risk could be managed and exhibit three presented information about how interest rate risk could be managed.

The initial requirement asked candidates to explain Fondir Co's risk exposure in the Italian market and to suggest ways in which this risk could be managed. Candidates were then asked to perform hedging calculations based on a foreign currency receipt from a customer in the country of Lothil and a short-term investment in US dollars. Candidates were then expected to discuss the results from their calculations and address associated queries raised by the company's board. The final requirement expected candidates to discuss if it would be beneficial for Fondir Co to manage its financial risks and whether or not the company should communicate its risk management approach to its stakeholders.

General comments

The requirements in this question are technically demanding. Overall, candidates performed adequately on this question, mainly due to the hedging calculations in requirement (b)(i) and discussion sections in requirements (b)(iii) and (b)(iv). However, the responses to requirement (a) on economic risk were quite mixed and the hedging calculations in requirement (b)(ii) were disappointing.

The best candidates were able to adopt a balanced approach to their answers. Once again, candidates are reminded about the importance of question practice under exam conditions as an essential part of their preparations for this exam. This type of preparation will also help candidates to learn to take the time to read the question and requirements carefully, whilst also learning to apply their knowledge to a specific scenario.

Part (a) – 6 marks

Requirement (a) asked candidates to explain the exposure Fondir Co faces with respect to the falling sales revenue from Italy and to suggest how this exposure could be managed. By highlighting the difficulties associated with managing certain risks using derivatives, exhibit one provided a strong hint to candidates that Fondir Co is exposed to economic risk.

Overall, this requirement was not answered well since many candidates did not identify the exposure to economic risk, wasting valuable exam time on transaction risk or unrelated business risks. Even many of those candidates who were able to identify economic risk being behind the decline in Fondir Co's revenue in its Italian market seemed to misunderstand the source of this risk, often confusing it with transaction risk and short term movements in the exchange rate when it is the impact on a company's cash flows caused by long-term macroeconomic factors. Many candidates did not recognise this long-term dimension and therefore incorrectly focussed on short-term hedging techniques in their suggestions for managing Fondir Co's risk exposure.

Part (b)(i) – 7 marks

Requirement (b)(i) asked candidates to calculate the amounts receivable in \$ from Lothil in four months' time, using the over-the-counter (OTC) forward rate and the OTC option.

Candidates were expected to use the data provided in exhibit two to calculate an appropriate forward rate based on interest rate parity theory. The most common error involved using annual interest rates to calculate the forward rate rather than adjusting for the four month period relevant to Fondir Co's expected receipt.

In the OTC option calculation, many candidates incorrectly identified this as a put option and/or ignored the need to add interest to the cost of the premium even though it is clear from the scenario that Fondir Co will borrow the money needed for the premium using its overdraft facility. In addition, a significant number of candidates wasted time calculating the number of contracts, fundamentally confusing an OTC option with an exchange-traded option.

Overall, this requirement was answered adequately although there were a few candidates who did not attempt it at all. It is important to attempt all requirements to gain maximum technical and professional marks.

Part (b)(ii) – 9 marks

This requirement expected candidates to calculate the interest return in dollars of the cash flows from Lothil when futures contracts are used to hedge the interest rate fluctuations of 0.5%. The requirement wording also explicitly instructed candidates to calculate the gain or loss in the futures market in dollars and the number of contracts that would be required. In order to answer this requirement, candidates were therefore expected to perform a number of detailed hedging calculations and to recognise the need for clear hedging instructions. Overall, the responses to this requirement were quite disappointing.

A significant number of candidates failed to identify whether the futures hedge is initiated by taking a long or short position and often ignored the instruction to calculate the number of contracts that would be required. Candidates therefore need to recognise the need for clear instructions when setting up a hedge. Those candidates who made it clear that they were buying 14 June contracts quickly earned two easy marks and seemed far more able to progress quickly and accurately through the rest of the calculations.

There were also errors with the unexpired basis and expected futures price calculations and future candidates would do well to focus on this area in their preparation for this exam. A small number of candidates used the lock-in rate as a shortcut method to calculate the annual return even though the requirement wording is clear that the gain or loss in the futures market and the interest return had to be calculated in dollars, rather than percentages. Such an approach severely restricted the credit that could be awarded to those candidates. A surprising number of candidates wasted valuable exam time assessing whether the contracts should be exercised or not even though it should have been obvious this was not relevant to a futures hedge.

Part (b)(iii) – 9 marks

This requirement expected candidates to comment on their results from parts (b)(i) and (b)(ii) and to address a query raised by the board in exhibit two with respect to the alternative methods which could be used to hedge the receipt and another query raised in exhibit three relating to the margin requirements associated with the futures hedge. There were therefore three aspects to this requirement.

Most candidates made a number of relevant points based on their results from parts (b)(i) and (b)(ii) with many recognising the flexibility afforded by an OTC option being at the expense of the option premium in addition to the comparative benefit of the OTC forward. However, few responses discussed the validity of the no basis risk assumption or the implication for the actual futures market outcome when discussing the results from part (b)(ii). The most common omission made by a significant number of candidates was ignoring the need for a comparison of the OTC forward and OTC option outcomes, which should have been a key aspect of their discussion in requirement (b)(iii)

A significant number of candidates ignored the part of the requirement relating to the alternative methods which could be used to hedge the lira receipt or wasted time suggesting alternative derivatives which had been specifically excluded by the board in the information provided in the scenario.

Most candidates presented relevant points in the discussion about the margin requirements, typically with a few sentences explaining the mechanics of an open futures position being marked-to-market daily. However, few candidates were able to expand on their discussion and address the board's specific query about whether or not dealing with futures is expensive due to the margin requirements associated with futures contracts.

Overall, this requirement was adequately answered although many candidates ignored one aspect of the requirement, most frequently the alternative methods which could be used to hedge the receipt. Candidates are therefore reminded to plan their response carefully and that no part of a question should be left unanswered.

Part (b)(iv) – 9 marks

This requirement asked candidates to discuss if it would be beneficial for Fondir Co to manage its financial risks and whether or not the company should communicate its risk management approach to its stakeholders. Exhibit one provided background information on the board's historic approach to risk management and its support for derivative products to be used to manage financial risk but also included a suggestion from the marketing director that the costs of risk management might outweigh the benefits.

The information within this exhibit therefore provided candidates with a useful framework within which to provide a well-structured response. Successful candidates took the time to read the exhibit and plan their answers carefully, directly addressing both aspects of the requirement.

Overall, this requirement was well answered by many candidates although the time constraint appeared to be a factor in a minority of cases. A number of candidates ignored or rushed the discussion relating to the benefits of risk management but were able to make relevant points justifying the proposed communication policy and its impact on the company's stakeholders, often focusing on the demands and importance of various stakeholder groups and how each of these would benefit from the new policy. However, very few responses considered the agency related costs, clientele effects or confidentiality issues associated with the communication policy.

Professional skills – 10 marks

As with the September 2022 session, it was pleasing to see that candidates seemed to be well prepared for the professional skills. That said, many candidates were more capable at demonstrating skills in communication and analysis and evaluation but were often less confident demonstrating skills in scepticism and commercial acumen.

Communication

Communication marks were available for:

- the general report format and structure
- the style, language and clarity of the report, including appropriate use of the available answer space options.
- the effectiveness of the communication and the ease with which it could be read

Many candidates scored well on the communication marks available.

Analysis and evaluation

Many candidates were able to demonstrate a reasonable level of skill in analysis and evaluation by providing comprehensive and well-structured numerical analysis for their calculations in their responses to requirements (b)(i) and (b)(ii). There was also credit available for a balanced appraisal and many candidates scored well here too.

Scepticism

To demonstrate the skill of scepticism, candidates were expected to provide an effective challenge to the information in the scenario. An example of effective scepticism might have involved questioning the assumption that the futures hedge is not subject to basis risk and identifying the implication for the futures outcome if this assumption is relaxed. Candidates seem to struggle demonstrating this particular skill and are advised to focus on this area in their preparation for the exam.

Commercial acumen

Few candidates were able to adequately demonstrate skills in commercial acumen, where candidates are expected to use the information in the scenario to draw evidence that relates to the organisational context or take any other practical commercial considerations into account.

Section B

Question 2 – Propleis Co

The screenshot shows the AFM December 2022 (22/23 syllabus) question interface. The top bar includes a title bar with the syllabus name and a window control icon, and a toolbar with icons for Symbol, Highlight, Strikethrough, Calculator, and Scratch Pad. The main area is divided into a left sidebar and a right content area. The sidebar contains three sections: Exhibits, Requirements, and Response Options. The Exhibits section lists two items: 1. Propleis Co and Adictcan Co and 2. Acquisition valuation. The Requirements section shows a single item: Requirements (25 marks). The Response Options section shows two items: Word Processor and Spreadsheet. The right content area contains the following text: "The following exhibits, available on the left-hand side of the screen, provide information relevant to the question: 1. Propleis Co and Adictcan Co 2. Acquisition valuation This information should be used to answer the question requirements within your chosen response option(s)." The bottom bar includes a Help/Formulae Street icon on the left and a navigation bar with Previous, Navigator, and Next icons on the right.

Format of the question

This question considered the potential acquisition by Propleis Co of another company in a similar business area. Initially candidates had to value the combined company and the benefits that could accrue to Propleis Co as a result of the acquisition. After this, candidates were asked to discuss assumptions made in the valuation and whether or not the expected synergies would be achieved. Finally, candidates were asked to explain actions that could be taken to ensure that the companies are integrated successfully.

General comments

This question was, generally, answered quite well. Inevitably errors and omissions were still made. These will be discussed within the individual parts of the question and future candidates should seek to avoid them.

In the valuation calculation it was good to see that the majority of candidates were able to make use of a well-structured approach that they had learned. Even though errors of detail were made candidates generally seemed to know what was required of them. The calculation of the benefits that could accrue to Propleis Co required calculations very specific to the scenario. Some candidates were able to flex their knowledge to suit the scenario and quickly carried out suitable calculations. Other candidates tried to apply a standard approach that they had learned without accounting for the particulars relevant in this question. This ability to flex your knowledge to the particular requirement in a question is a key skill required in AFM, where the aim is to test your ability to apply your knowledge to any given scenario.

In order to develop this skill candidates preparing for this exam should aim to work as many past exam questions as possible.

When discussing assumptions made candidates need to make sure that they do actually discuss the assumptions. Too many candidates simply state assumptions which are very often given to them in the scenario anyway. The credit available for this is obviously limited.

Candidates should recognise that an answer which comprises a general discussion of related points will always earn fewer marks than an answer which really tries to tackle the question posed in the requirement. Furthermore, answers which do tackle the actual question posed are far more likely to earn the professional skills marks available.

Many responses to the discursive areas were rather brief and hence could never be expected to earn all the marks available. As a general rule candidates should try to make one point per mark available although there is quite often up to two marks available for a point that is well developed. Time pressure did not seem to be a major issue in this question and hence the brevity of the responses is most likely to have been due to candidate uncertainty or lack of knowledge.

Part (a) – 10 marks

Estimate:

- **the equity value of the combination of Propleis Co and Adictcan Co; and**
- **the benefits which could be gained by Propleis Co's Shareholders from the acquisition**

The first element of this question part was quite well done. The vast majority of candidates calculated the sales and post-tax operating cash flows for the first four years correctly. Furthermore, most candidates also calculated the additional capital investment correctly although more errors arose here. When doing any calculation candidates should always try to think what a sensible answer would be. In this question the additional capital investment in year one was given and was \$25m. Hence when calculating the figure for year two a similar figure should be expected – here it was \$17m. If a candidate's calculation results in a markedly different figure such as \$1.2m or \$104m the candidate would be wise to check their calculations as they could have made an error. The additional capital investment was then subtracted from the post-tax operating cash flows to give the free cash flows for the first four years. Even here some unnecessary errors arose as, for instance, the sales figures were sometimes added into the free cash flow calculation or the additional capital investment was added rather than subtracted. The free cash flows

for the first four years were then discounted at the discount rate given to produce the PV of the first four years of cash flows.

The PV of the delayed perpetuity from Year 5 then needed to be calculated. This is something that has been frequently examined over the years and candidates would be well advised to ensure they can do it.

The steps are as follows:

1. Calculate the cash flow expected in year five by inflating the prior year cash flow by the expected growth rate:

$$\$107\text{m} \times 1.04 = \$111.3\text{m}$$

2. Multiply the cash flow expected by the perpetuity with growth formula ($1/(r-g)$):

$$\$111.3\text{m} \times 1/(0.1 - 0.04) = \$1855\text{m}$$

3. Recognise that the perpetuity is delayed. The perpetuity with growth formula assumes that the first cash flow arises at T1 whereas here the first cash flow arises at T5. Hence it is delayed by 4 years and must be multiplied by the 4-year discount factor to get the PV:

$$\$1855\text{m} \times 0.683 = \$1267\text{m}$$

The biggest mistake that candidates make is to fail to do step 3 or to use the 5-year discount factor in step 3 presumably because the perpetuity starts at T5. Once you are confident with this calculation there is no need to do it step by step but until you are confident you should work through it logically.

The total value of the company could then be calculated by adding the PV of the cash flows for the first 4 years to the PV of the perpetuity from Year 5. Candidates needed to recognise that this total value includes the value of both the equity and the debt within the company. To calculate the total equity value the target equity to debt ratio needed to be used. Candidates should recognise that in a future question they may be using free cash flows to equity (post interest paid) and discounting at the cost of equity. In this case the value calculated is the total equity value.

The second element of this question part asked candidates to estimate the benefits that would be gained by the Propleis Co's shareholders from the acquisition. There are a number of ways candidates could have done this but perhaps the simplest way would be as follows:

Value calculated less the current value of both companies = total synergies available.

Total synergies available less the gain required by the Adictcan Co shareholders = gain available to the Propleis Co shareholders.

This gain available could then be converted into a % gain to compare against the % target set.

Unfortunately, many candidates seemed to lack the confidence to proceed or if they did proceed tried to apply what they had seen in other questions, splitting the synergy gain equally between the two companies for instance, rather than using the information given to them in the scenario.

Part (b) – 6 marks

Discuss the assumptions made in the calculations in (a), including whether or not the expected synergies are likely to be achieved.

Candidates proved to be good at listing assumptions, many of which were given to them in the question, but much less good at really discussing the assumptions.

For example, a candidate might say 'it is assumed that after 4 years the free cash flows will grow at a rate of 4% per year'. This is given in the question and hence can only earn minimal merit. Much more merit is available if the assumption is properly discussed - as the model answer says it may be unrealistic to assume that the magazines will continue to create cash in perpetuity even though they have a long history.

Candidates who generate assumptions not specifically given in the scenario are likely to earn more credit. For instance, in this question it has effectively been assumed that depreciation is equal to replacement asset investment. Candidates who then discussed this further by stating that this may be optimistic as depreciation is based on past costs, whereas replacement asset investment incurs current day costs, which are likely to be higher, would earn even more credit.

As well as discussing whether or not assumptions made may be reasonable candidates should go one step further and identify where it would be useful to have further analysis or information. In this question it would, for instance, be useful to have more information on how the cost of capital was calculated and hence whether or not it can be justified. Equally candidates should consider the implication if some of the assumptions were to prove inaccurate. The basic implication is that the value and gains calculated could be either under or over stated which could obviously impact on any decision regarding the acquisition. Unfortunately, only a few candidates get this far. Candidates who do take this extra step tend to earn more of the marks available.

Candidates' answers to whether or not the expected synergies were likely to be achieved were also disappointing to some extent. Too many scripts focused on describing the synergies that might arise and only made a passing reference, if any, to whether or not they could be achieved. Candidates should always be thinking about how they could structure their answer to best answer the actual question posed. In this question part candidates could have had a couple of sub-headings, 'Synergies likely to be achieved' and 'Synergies less likely to be achieved'. Under 'Synergies likely to be achieved' candidates could, for instance, have discussed the ability to bulk buy paper or save in administration cost as both companies will probably need similar paper and administration. Under the heading 'Synergies less likely to be achieved' candidates could have discussed that cross selling may be limited as the magazines of the two companies appeal to very different types of readers. Candidates who try to structure their answers in this way tend to earn more of the technical marks available as they are much better placed to answer the question posed and also tend to earn more of the professional skills marks available.

Part (c) – 4 marks

Explain the actions which Propleis Co's board can take to ensure that the companies are integrated successfully and synergies are realised.

In general candidates were better at answering the question posed here when compared with the discussion of whether or not the expected synergies would be achieved in part (b). Indeed, some candidates earned very good marks.

Common issues in candidates' answers included the following:

- Discussing only a very limited number of actions that the board could take. Many candidates focused on the need to avoid a clash of cultures arising and the need for good communication within the newly merged firm but little else. In general, it is better to make more succinct points than it is to go into great depth on just a couple of areas.
- Candidates who had discussed synergies generally in part (b) often started to discuss whether or not they would be achieved in this question part. Where this occurred, they were given suitable credit in part (b).
- Discussing the actions too generally and not relating their discussion to the scenario in the question.

Professional skills – 5 marks

Again many candidates were more capable at demonstrating skills in analysis and evaluation but were often less confident demonstrating skills in scepticism and commercial acumen.

Analysis and evaluation

Many candidates were able to demonstrate a reasonable level of skill in analysis and evaluation by providing comprehensive and well-structured numerical analysis for their calculations in part (a) and using this analysis to support their discussion in parts (b) and (c).

Scepticism

To demonstrate the skill of scepticism, candidates were expected to provide an effective challenge to the information in the scenario. An example of effective scepticism might have involved challenging the accuracy of the forecast information and its impact on the final valuation.

Commercial acumen

Few candidates were able to adequately demonstrate skills in commercial acumen, where candidates are expected to use the information in the scenario to draw evidence that relates to the organisational context or take any other practical commercial considerations into account.

Question 3 – Tonpantau Co

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Symbol Highlight Strikethrough Calculator Scratch Pad Close All Flag for Review

Exhibits

- 1. Tonpantau Co proposed investment
- 2. Details of investment

Requirements

- Requirements (25 marks)

Response Options

- Word Processor
- Spreadsheet

The following **exhibits**, available on the left-hand side of the screen, provide information relevant to the question:

1. Tonpantau Co proposed investment
2. Details of investment

This information should be used to answer the question **requirements** within your chosen **response option(s)**.

Help | Formulae | Street < Previous Navigator Next >

Format of the question

In this question, candidates were asked to consider a publishing company which currently produces mathematics and business studies material for schools but was thinking about producing material for university courses and professional qualifications. An appraisal of the proposed investment was required and then candidates were required to discuss some aspects of real options with reference to the proposed investment.

General comments

This question was, generally, answered quite well. Inevitably errors and omissions were still made. These will be discussed within the individual parts of the question and future candidates should seek to avoid them.

Initially candidates needed to calculate a WACC to be used as the discount rate in the appraisal. In order to do this both a cost of equity and a cost of debt had to be calculated. Whilst the cost of equity calculations were often done very well, the cost of debt calculations were often poorly executed or completely missing. Having calculated the WACC candidates then needed to calculate the NPV of the proposed investment. The vast majority of candidates were able to make a fair attempt at the NPV calculation. Candidates were able to complete all the calculations required of them using standard approaches that they should have seen in many previous questions and which they should have committed to memory. Indeed, a good number of candidates who had done this were able to earn full marks for the calculations in this question.

With regard to the real options discussion candidates were quite good at describing how real options build on traditional NPV and were able to identify a good number of

problems associated with incorporating and valuing real options using the Black-Scholes model. However, candidates' discussion tended to be rather generic and did not make much use of the scenario which they had been provided with. In particular many candidates failed to discuss the real options apparently available to Tonpantau Co which rather reduced the marks that they could gain. Additionally, whilst candidates who generate only generic points can still earn some technical marks their ability to earn the available professional skills marks is much reduced.

Many responses to the discursive area were rather brief and hence could never be expected to earn all the marks available. As a general rule candidates should try to make one point per mark available although there is quite often up to two marks available for a point that is well developed. Time pressure did not seem to be a major issue in this question and hence the brevity of the responses is most likely to have been due to candidate uncertainty or lack of knowledge. Candidates who made better use of the scenario were very often in a better position to write a fuller answer.

Part (a) – 12 marks

Evaluate the financial acceptability of the proposed investment.

This question part could be broken down into several discrete calculations:

Cost of equity:

In order to calculate the cost of equity candidates needed to de-gear a proxy company beta and then re-gear it using the debt/equity ratio of Tonpantau Co. Finally, the cost of equity could be calculated by using the re-gear proxy beta in the capital asset pricing model formula. These calculations were often done very well indeed. The most common mistakes were as follows:

- De-gearing and/or re-gearing using the wrong debt/equity ratios.
- Failing to recognise the need to re-gear to reflect the financial risk of Tonpantau Co.
- Treating the risk premium provided as if it was the return on the market.

Cost of debt:

The necessary calculations here were often poorly done or were not done at all. Initially candidates needed to value the bond using the spot rates given in the question. Candidates who did this often did it well although errors such as these still arose:

- Failing to recognise that the spot rates given are annual rates. For instance, the annual spot rate for 2 years was given as 5.15% and hence the 2-year discount factor must be calculated as follows – $1/1.0515^2$.
- Treating the spot rates as if they were the coupon and discounting at the coupon rate. This is effectively the reverse of what was necessary.

Having calculated the value of the bond candidates then needed to calculate the cost of the bond, and hence the cost of debt, by calculating the IRR of the relevant cash flows. Unfortunately, a number of candidates who successfully calculated the value of the bond did not then know how to proceed. The relevant cash flows are the value of the bond just calculated, the annual interest on the bond and the redemption value of the bond. A significant number of candidates neglected the premium on redemption and/or the final interest payment on the bond. Candidates who were able to use the IRR function within the spreadsheet provided were able to complete their calculations quickly and accurately and hence earn marks quickly. All candidates should aim to become confident in using the spreadsheet functions available to them as using them saves considerable time and greatly reduces the chance of minor calculative errors.

WACC:

Having calculated a cost of equity and a cost of debt candidates were then in a position to calculate the WACC. The main error that arose here was that some candidates failed to account for the tax benefit of the debt finance when calculating the WACC.

NPV:

Having calculated the WACC to be used as the discount rate in the NPV calculation candidates then needed to construct a cash flow table for the proposed project. The vast majority of candidates had a reasonable structure that they could use here. The better structures made a clear distinction between revenue cash flows and capital cash flows. Those that fail to do this are always at risk of applying revenue tax to capital cash flows which is not correct.

Common errors that arose in the cash flow table were as follows:

- Failing to recognise that pre-tax profits provided were after the deduction of the tax-allowable depreciation (TAD).
- Delaying the initial working capital investment until T1.
- Failing to calculate the incremental working capital cash flows from the total working capital required that was given.

- Treating all the capital investment as if it took place at T0.

From the cash flow table, the net cash flows could then be calculated and discounted at the WACC to give the NPV, which in this case was negative. It was good to see that a majority of candidates were using the NPV function available in the spreadsheet. Candidates were expected to state that a negative NPV means that the project should be rejected as it will not add to shareholder wealth. As ever some candidates failed to do this and missed out on marks that they could have very easily gained.

Further points on calculative question parts:

Candidates preparing for this examination should be working past questions to time in proper exam conditions. They should then be comparing their results with the model answer in order to discover where they have gone wrong. This can be very dispiriting as inevitably errors are made and the answers calculated can be very different from the model answer. Candidates should be aware that answers which may look very wrong can still earn very good marks. In the question above a candidate who made a mistake in their cost of equity calculation, their cost of debt calculation, and in their cash flows would have ended up with very different answers to those in the model answer. However, you are only penalised once for each error and hence an answer such as this could still earn up to 10 of the 13 technical marks available. Candidates who persevere with this approach when studying are far more likely to be successful when they sit the exam, as they will have learned how to overcome the sort of issues that arise in AFM questions.

Candidates who get stuck at some point in their calculations must not be afraid of making assumptions and then carrying on. As previously stated, a significant number of candidates calculated the value of the bond but could not then calculate the cost of the bond. Those with good exam technique simply stated an assumed cost of debt and proceeded to calculate a WACC using this. In this way they only lost 2 technical marks. Other candidates let their calculations falter, they failed to calculate a WACC and did not discount the cash flows to arrive at a NPV as they didn't have a WACC. Candidates who did this lost at least 5 technical marks.

Now that the exams are computer-based candidates have use of a spreadsheet to assist with their calculations. Some candidates are still doing all their calculative work without using the spreadsheet at all which wastes a lot of time and is rarely successful.

Whilst the spreadsheets are a useful tool for candidates, they do mean that some new errors/problems are arising. Common errors/problems that have been noticed are as follows:

- Candidates may, for instance, show a tax cost correctly as a negative in a cash flow table and then when adding up the column or row will deduct the tax. However, if you deduct a negative figure the result is that the figure is added. Hence the tax cost is finally treated incorrectly.
- Candidates add up columns in a cash flow and include rows which should not be included. For instance, a question may require you to estimate sales and from that you can calculate the estimated operating cash flows. Candidates then forecast the other cash flows required for each year and when adding the columns prior to discounting they mistakenly add in both the sales and the operating cash flows. This is obviously wrong but is easily done when working at speed in exam conditions.
- Some candidates try to calculate too much in one cell of the spreadsheet and sometimes a cell may have a long calculation which could be worth a number of marks. Errors are far more common than if the calculation was broken down into bite size chunks. Furthermore, when long calculations like this are used it is much harder for a marker to see what has been done well and give the appropriate credit.

Candidates should try to prevent these errors/problems from creeping into their work.

Part (b) – 8 marks

Discuss, with reference to Tonpantau Co's proposed investment:

- **How real options build on traditional net present value analysis when evaluating investment decisions; and**
- **The problems with incorporating and valuing real options. Your answer should make specific reference to the Black-Scholes model.**

A minority of candidates were unable to make any real attempt at this question part as they had minimal or no knowledge of real options in investment appraisal. This indicates a lack of exam preparation.

The initial element of this question part was reasonably well answered. Many candidates were able to explain that the flexibility inherent in projects could add value as it enables companies to react to uncertain future events in order to maximise their return from a project. Candidates often mentioned the now or never nature of traditional NPV analysis, which assumes a project in its entirety is started now and must be completed as currently anticipated and hence ignores any flexibility which may exist. Candidates often also correctly identified that traditional NPV effectively accounts for intrinsic value but not the time value of options that may be available. Candidates need to be careful how they word their answers to ensure their

meaning is clear. In this question a significant number of candidates simply stated that NPV ignores time value which is obviously not correct.

It was hoped that candidates would then refer to and discuss the real options available to Tonpantau Co which were; the option to withdraw or abandon if sales proved disappointing, and the option to expand into producing material for other university courses and professional qualifications. Although these were clearly signposted in the scenario, many candidates did not refer to them at all and instead simply produced a generic list of potential real options with no reference to the scenario.

Those who discussed the two options apparently available to Tonpantau Co could earn up to 4 technical marks. An extra technical mark was also available for any candidate who came up with any other real option that Tonpantau Co could benefit from. For instance, if the expansion into producing material for mathematics degree courses was successful the company could then produce material for business studies degree courses. Very few candidates earned many of these technical marks.

The second element of this question part, discussing the problems with incorporating and valuing real options, was answered much better. However, once again candidates' answers tended to be rather generic and did not use the scenario as much as had been hoped for. Despite this many candidates could identify many relevant problems and earned good marks here. Indeed, many candidates wrote more than was necessary for the technical marks available.

Professional skills – 5 marks

As with question 1, many candidates were more capable at demonstrating skills in analysis and evaluation but were often less confident demonstrating skills in scepticism and commercial acumen.

Analysis and evaluation

Many candidates were able to demonstrate a reasonable level of skill in analysis and evaluation by providing comprehensive and well-structured numerical analysis for their calculations and drawing appropriate conclusions in part (a).

Scepticism and commercial acumen

Candidates rarely earned many professional skills marks under these categories as these marks generally require candidates to be making good use of the scenario and this is something that fewer candidates did well. It is hoped that candidates' performance in this area will improve as experience with the professional skills marks grows.